

Reaction–diffusion and Turing patterns

How diffusion can destabilize a uniform membrane

Diffusion smooths each concentration field in isolation. Coupled to reactions, unequal diffusion can instead destabilize a homogeneous state. The essential calculation is a competition between the reaction Jacobian and the spatial Laplacian spectrum. This note starts with two species on a fixed, locally flat membrane.

1 Separate local chemistry from spatial transport

Let $u(\mathbf{x}, t)$ and $v(\mathbf{x}, t)$ be surface concentrations, with reaction rates $f(u, v)$ and $g(u, v)$ and constant positive diffusivities D_u, D_v :

$$\partial_t u = f(u, v) + D_u \nabla^2 u, \quad \partial_t v = g(u, v) + D_v \nabla^2 v. \quad (1)$$

Choose a homogeneous steady state (u_0, v_0) with $f = g = 0$. All derivatives below are evaluated there. Its local reaction Jacobian is

$$J = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}, \quad \text{tr } J = f_u + g_v < 0, \quad \det J = f_u g_v - f_v g_u > 0. \quad (2)$$

These inequalities make the well-mixed state linearly stable. They are essential: a spatial pattern developing from an already unstable local reaction system need not be a Turing instability.

A perturbation proportional to $e^{st+i\mathbf{q}\cdot\mathbf{x}}$ evolves with

$$J_q = J - q^2 \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}, \quad q = |\mathbf{q}|. \quad (3)$$

Since $\text{tr } J_q = \text{tr } J - (D_u + D_v)q^2 < 0$, instability requires the determinant to become negative:

$$\det J_q = \det J - (D_v f_u + D_u g_v)q^2 + D_u D_v q^4. \quad (4)$$

Thus the instability is stationary at onset in this two-species, diagonal-diffusion model; it is not a finite-wavenumber Hopf bifurcation.

2 The classical Turing conditions

The quadratic polynomial in q^2 in Eq. (4) has a negative minimum at positive q^2 precisely when

$$D_v f_u + D_u g_v > 0, \quad (D_v f_u + D_u g_v)^2 > 4D_u D_v \det J. \quad (5)$$

Together with Eq. (2), these are the continuum-wavenumber conditions for a diffusion-driven instability. If $D_u = D_v = D$, then $J_q = J - Dq^2 I$: every growth rate shifts left, and the instability is impossible. Cross-diffusion or additional species can change this conclusion.

3 An example and a geometry test

In dimensionless units, take a local linearization and diffusion matrix

$$J = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \quad D_u = 1, \quad D_v = 10. \quad (6)$$

Then $\text{tr } J = -2$, $\det J = 1$, and

$$\det J_q = 1 - 7q^2 + 10q^4 = (1 - 5q^2)(1 - 2q^2). \quad (7)$$

The unstable band is therefore $1/5 < q^2 < 1/2$. For example, $q^2 = 0.3$ gives $\det J_q = -0.2$ and one positive growth rate. These numbers specify a linear example; nonlinear reaction terms are still needed to determine the final pattern and keep concentrations physical.

A finite domain does not permit arbitrary q . On a periodic one-dimensional interval of length L , $q_n = 2\pi n/L$; at least one allowed nonzero mode must enter the unstable band. Local kinetics satisfying Eq. (5) can therefore remain stable in a domain too small to fit an unstable mode.

On a fixed spherical membrane of radius R , replace the planar Laplacian by the Laplace–Beltrami operator. Spherical harmonics satisfy

$$-\nabla_S^2 Y_{\ell m} = \frac{\ell(\ell+1)}{R^2} Y_{\ell m}, \quad \ell = 0, 1, 2, \dots \quad (8)$$

The same determinant test applies with $q^2 = \ell(\ell+1)/R^2$. Radius selects which angular modes can grow. Curvature enters through this mode spectrum in the present model; curvature-dependent chemistry or membrane deformation would add further couplings.

4 The biological model determines the conservation law

Local self-activation and a faster-spreading inhibitory influence provide a common realization of the inequalities. But not every intracellular pattern is a classical two-species activator–inhibitor pattern. A particularly important distinction is whether the total amount of protein is conserved.

If the two species interconvert with $g = -f$, then

$$\partial_t(u + v) = D_u \nabla^2 u + D_v \nabla^2 v, \quad \frac{d}{dt} \int (u + v) dA = 0 \quad (9)$$

under no-flux boundaries. The homogeneous Jacobian has $\det J = 0$: the conserved total produces a neutral uniform mode, so the strict stability conditions in Eq. (2) are no longer the right starting point. Spatial redistribution can still destabilize the system, but must be analyzed at fixed total mass [2].

Actual membrane–cytosol exchange couples surface and volume concentrations and often involves more than two chemical states. A two-dimensional surface model is a reduction whose assumptions should be checked. Linear instability predicts early growth and admissible wavelengths; it does not by itself select spots, stripes, traveling waves, or a unique final state.

5 A complete reaction model: Schnakenberg kinetics

The arbitrary Jacobian in the overview shows that instability is possible. A nonlinear reaction model also tells us where the Jacobian comes from. Consider the dimensionless Schnakenberg activator–substrate system [3]

$$\partial_t u = a - u + u^2 v + \nabla^2 u, \quad (10)$$

$$\partial_t v = b - u^2 v + D_v \nabla^2 v, \quad a, b > 0. \quad (11)$$

Time is measured in the activator decay time, length in its diffusion length, and the concentration scaling sets the autocatalytic coefficient to one. Thus $D_u = 1$ and D_v is a ratio of physical diffusivities. The reactions can be understood as constant supply of both species, linear removal of u , and autocatalytic conversion of substrate into activator. They need not represent a particular membrane protein network.

Adding the steady-state equations gives $u_0 = a + b$. Substituting into $g = 0$ then gives $v_0 = b/(a + b)^2$. Differentiating the reaction terms explicitly yields

$$J = \begin{pmatrix} (b - a)/(a + b) & (a + b)^2 \\ -2b/(a + b) & -(a + b)^2 \end{pmatrix}, \quad \det J = (a + b)^2. \quad (12)$$

The determinant is positive, while local stability additionally requires $(b - a)/(a + b) < (a + b)^2$. Choose $a = 0.1$ and $b = 0.9$:

$$(u_0, v_0) = (1, 0.9), \quad J = \begin{pmatrix} 0.8 & 1 \\ -1.8 & -1 \end{pmatrix}, \quad \text{tr } J = -0.2, \quad \det J = 1. \quad (13)$$

The determinant for a Fourier mode is

$$\det J_q = 1 - (0.8D_v - 1)q^2 + D_v q^4. \quad (14)$$

The diffusion-driven instability needs $0.8D_v - 1 > 0$ and $(0.8D_v - 1)^2 > 4D_v$. On the branch with positive $0.8D_v - 1$, the threshold is

$$D_v > D_{v,c} = \frac{5.6 + \sqrt{28.8}}{1.28} \simeq 8.568. \quad (15)$$

At $D_v = 10$ the unstable interval is again $0.2 < q^2 < 0.5$, but the trace and therefore the growth rates differ from the earlier arbitrary-Jacobian example. The determinant fixes the band edges, not the full time evolution.

The unstable eigenvector describes relative concentrations

For a mode with growth rate s , the first row gives

$$\frac{\delta v}{\delta u} = \frac{s - f_u + D_u q^2}{f_v}. \quad (16)$$

At $D_v = 10$ and $q^2 = 0.3$, the positive eigenvalue is about 0.05624, so $\delta v/\delta u \simeq -0.44376$. Activator-rich regions are substrate-poor in this mode. This sign is a model prediction about the pattern's composition, not merely its wavelength.

6 Dispersion, fastest growth, and finite geometry

Define $T(q) = \text{tr } J_q$ and $\Delta(q) = \det J_q$. The two eigenvalues are

$$s_{\pm}(q) = \frac{T(q) \pm \sqrt{T(q)^2 - 4\Delta(q)}}{2}. \quad (17)$$

If the discriminant is negative, both modes have real part $T/2 < 0$ and decay while oscillating. Inside a Turing band, $\Delta < 0$ gives two real eigenvalues of opposite sign.

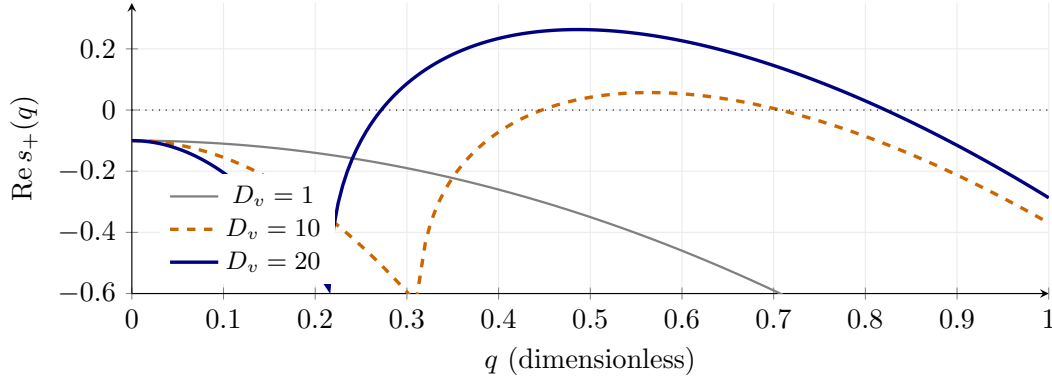


Figure 1: Exact linear dispersion for the Schnakenberg example with $a = 0.1$, $b = 0.9$, and $D_u = 1$. Increasing the diffusion ratio creates and widens a band of growing modes.

To distinguish the minimum determinant from the fastest mode, write $y = q^2$ and differentiate the characteristic equation $s^2 - Ts + \Delta = 0$:

$$\frac{ds}{dy} = \frac{T's - \Delta'}{2s - T}. \quad (18)$$

At an interior maximum of the unstable branch, $\Delta' = T's = -(D_u + D_v)s$. The minimum of Δ , where $\Delta' = 0$, therefore coincides with fastest growth only at threshold when $s = 0$. Away from onset, minimizing the determinant gives the wrong growth-maximizing wavenumber.

For $D_v = 10$, the first periodic mode in a domain of length $L = 10$ has $q_1^2 = (2\pi/10)^2 \simeq 0.395$ and lies in the unstable band. The second mode is outside it. At $L = 5$ no nonzero mode is unstable. At $L = 20$, the second mode is unstable while the first is below the band. These checks use dimensionless lengths consistent with the reaction equations.

On a sphere, insert $q^2 = \ell(\ell + 1)/R^2$ into the same dispersion. The $2\ell + 1$ spherical harmonics at fixed ℓ are degenerate in this isotropic linear model. Linear theory selects an angular scale but not a unique orientation or nonlinear pattern. Boundary heterogeneity, nonlinear mode coupling, and noise can remove that ambiguity.

7 Mass redistribution has a different long-wavelength limit

For interconversion, write $\partial_t u = f(u, v) + D_u \nabla^2 u$ and $\partial_t v = -f(u, v) + D_v \nabla^2 v$. Let $n = u + v$ be the local total density. The reaction Jacobian has eigenvalues 0 and $f_u - f_v$. Assume $f_u - f_v < 0$, so reactions relax rapidly toward a local equilibrium at fixed n .

Let that equilibrium be $u_{\text{eq}}(n)$ and $v_{\text{eq}}(n) = n - u_{\text{eq}}(n)$. Differentiating $f(u_{\text{eq}}, n - u_{\text{eq}}) = 0$ gives

$$(f_u - f_v) \frac{du_{\text{eq}}}{dn} + f_v = 0, \quad \frac{du_{\text{eq}}}{dn} = \frac{f_v}{f_v - f_u}, \quad \frac{dv_{\text{eq}}}{dn} = -\frac{f_u}{f_v - f_u}. \quad (19)$$

At long wavelengths, reactions keep the local composition close to this curve. The total diffusive current becomes

$$\mathbf{J}_n \simeq -D_{\text{eff}} \nabla n, \quad D_{\text{eff}} = \frac{D_u f_v - D_v f_u}{f_v - f_u}. \quad (20)$$

If $D_{\text{eff}} < 0$, an increase in total density shifts the local equilibrium composition so that the resulting transport brings in still more mass. This is a mass-redistribution instability. Its connection to MIPS is negative collective diffusion after a fast variable is eliminated; the microscopic sources of feedback are different.

The exact two-field determinant is

$$\det J_q = (D_u f_v - D_v f_u) q^2 + D_u D_v q^4. \quad (21)$$

When $D_u f_v - D_v f_u < 0$, the unstable band extends from arbitrarily small nonzero q to a finite upper edge. Unlike the classical Turing case, $q = 0$ stays neutral because the total mass is conserved; it is not an exponentially growing uniform perturbation at fixed total mass.

Checkable example. Take $f_u = 1$, $f_v = 2$, $D_u = 1$, and $D_v = 10$ in consistent dimensionless units. Local reactions at fixed n decay with rate 1, but $D_{\text{eff}} = -8$ and $\det J_q = -8q^2 + 10q^4$. Hence $0 < q^2 < 0.8$ is unstable. The full two-field diffusion stabilizes large q without inserting an additional phenomenological fourth derivative by hand.

What remains after a linear calculation

Specify boundary conditions and total mass. Check a small Fourier perturbation against its predicted growth rate and eigenvector, then refine the grid and time step. Nonlinear saturation involves derivatives beyond the Jacobian; linear instability does not establish a stable patterned attractor.

Membrane–cytosol exchange couples densities with different units and geometries; reducing it to two surface fields needs justification. Synthesis, degradation, or reservoirs break conservation and can change both the neutral mode and the pattern scale.

References

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